DOMINATION OF A MAP IS EQUIVALENT TO UNIVERSAL INJECTIVITY OF THE PUSHOUT

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement contains an Xy-pic diagram; the preamble spans of this card run 8-18 of the chapter's own preamble file and therefore include the Xy-pic package and option setup at lines 15-18.

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Allowed standard prerequisites (editorial).

- The pushout as a quotient of a direct sum. Recalled at the start of the counted proof (lines 21166-21167). Gives the concrete description of the pushout module N' used to identify kernels.
- Tensoring commutes with pushouts. Used in the middle of the counted proof (lines 21173-21179). Produces the kernel sequence after tensoring with an arbitrary module, which is what compares the two conditions.

Lemma 0.1. *Let $f : M \to N$ and $g : M \to M'$ be maps of R-modules. Consider the pushout of f and g,*

$$\begin{array}{ccc} M & \xrightarrow{f} & N \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g'} \\ M' & \xrightarrow{f'} & N' \end{array}$$

Then g dominates f if and only if f' is universally injective.

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Proof. Recall that N' is $M' \oplus N$ modulo the submodule consisting of elements $(g(x), -f(x))$ for $x \in M$. From the construction of N' we have a short exact sequence

$$0 \to \mathrm{Ker}(f) \cap \mathrm{Ker}(g) \to \mathrm{Ker}(f) \to \mathrm{Ker}(f') \to 0.$$

Since tensoring commutes with taking pushouts, we have such a short exact sequence

$$0 \to \mathrm{Ker}(f \otimes \mathrm{id}_Q) \cap \mathrm{Ker}(g \otimes \mathrm{id}_Q) \to \mathrm{Ker}(f \otimes \mathrm{id}_Q) \to \mathrm{Ker}(f' \otimes \mathrm{id}_Q) \to 0$$

for every R-module Q. So f' is universally injective if and only if $\mathrm{Ker}(f \otimes \mathrm{id}_Q) \subset \mathrm{Ker}(g \otimes \mathrm{id}_Q)$ for every Q, if and only if g dominates f. $\square$